GLOBAL TUBERCULOSIS EPIDEMIOLOGICAL ANALYSIS

Analysis of Tuberculosis Incidence, Mortality, Cases, and Case Fatality Rates Across Countries and WHO Regions

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Field

Epidemiology & Healthcare Data Analytics

Data Source

World Health Organization (WHO)

Analytical Tools

Python | Pandas | Matplotlib | Seaborn | Power BI

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Executive Summary

Tuberculosis (TB) remains a major global public health challenge, with substantial differences in disease burden, incidence, mortality, and case fatality across countries and WHO regions. This project applies epidemiological and healthcare data analytics methods to examine the global distribution and patterns of tuberculosis using data obtained from the World Health Organization (WHO).

The analysis was conducted using Python for data exploration, data cleaning, feature engineering, statistical analysis, and visualization. The final cleaned analytical dataset contained 5,347 observations and 15 variables. Key indicators included total TB cases, TB deaths, incidence rate, mortality rate, and case fatality rate (CFR).

The analysis identified a total of approximately 281.4 million TB cases) and 49.9 million TB deaths across the analyzed observations. India recorded the highest cumulative number of TB cases (80.32 million) and TB deaths (14.15 million). The South-East Asia (SEA) WHO region recorded the highest cumulative TB case burden, with approximately 100.5 million cases.

Temporal analysis demonstrated substantial variation in TB cases and mortality across the study period, with the highest annual number of cases recorded in 2005, at approximately 2.05 million.

An interactive Power BI dashboard was developed to communicate the findings through key performance indicators, temporal trends, country-level comparisons, regional analysis, geographic visualization, and epidemiological relationships between TB incidence and mortality.

Overall, the findings demonstrate substantial geographic variation in the global TB burden and highlight the importance of strengthening early detection, treatment coverage, surveillance systems, and data-driven resource allocation, particularly in high-burden settings.

Introduction

Tuberculosis (TB) is an infectious disease caused primarily by Mycobacterium tuberculosis and remains one of the most important communicable diseases affecting global health. Although TB is preventable and treatable, it continues to cause substantial morbidity and mortality, particularly in countries and populations experiencing a high burden of disease.

The epidemiological distribution of tuberculosis is not uniform. TB incidence, mortality, treatment outcomes, and case fatality vary considerably across countries and geographic regions. These differences may reflect variations in healthcare access, diagnostic capacity, treatment coverage, socioeconomic conditions, population characteristics, comorbidities, and the effectiveness of national TB control programs.

Effective TB control requires reliable epidemiological surveillance and continuous monitoring of key indicators. Measures such as TB incidence, mortality, number of cases, number of deaths, and case fatality rate provide important information for assessing disease burden and identifying populations and geographic areas that require targeted interventions.

The increasing availability of health data provides opportunities to strengthen epidemiological analysis through data analytics and visualization. However, raw health datasets often require systematic cleaning, transformation, validation, and analysis before they can be effectively used for public health decision-making. Combining epidemiological methods with analytical tools such as Python and Power BI can facilitate the identification of temporal trends, geographic differences, and relationships between important disease indicators.

This project therefore focuses on the epidemiological analysis of global tuberculosis data obtained from the World Health Organization (WHO). The analysis examines TB cases, deaths, incidence rates, mortality rates, and case fatality rates across countries and WHO regions over time. The project also develops an interactive Power BI dashboard to communicate the findings and support data-driven interpretation of the global TB burden.

Problem Statement

Tuberculosis remains a significant public health concern despite the availability of effective diagnostic and treatment interventions. The burden of TB is unevenly distributed across countries and regions, with substantial differences in the number of cases, deaths, incidence rates, mortality rates, and case fatality rates.

One of the major challenges in TB surveillance is the ability to transform large and complex epidemiological datasets into clear, interpretable information that can support timely public health decisions. Raw datasets may contain missing values, inconsistent variables, different measurement scales, and multiple observations across countries and years, making direct interpretation difficult.

Furthermore, differences in TB burden between countries and WHO regions require systematic analysis to identify high-burden settings and areas with elevated mortality. Understanding temporal trends and the relationship between incidence and mortality is also important for evaluating disease patterns and identifying areas where additional public health interventions may be required.

Therefore, there is a need for a structured epidemiological data analysis approach that integrates data cleaning, statistical exploration, visualization, and interactive reporting. Such an approach can improve understanding of the global TB burden and facilitate evidence-based decision-making.

This project addresses this problem by analyzing WHO tuberculosis data using Python and developing an interactive Power BI dashboard. The analysis aims to identify geographic and temporal patterns in TB cases, deaths, incidence, mortality, and case fatality, while presenting the findings in an accessible format for epidemiological interpretation and public health planning.

Objectives

1/ General Objective

To analyze global tuberculosis epidemiological patterns using WHO data and develop an interactive data analytics dashboard to support evidence-based public health decision-making.

2/ Specific Objectives

The specific objectives of this project are to:

1. Assess the global burden of tuberculosis by examining the distribution of TB cases and deaths across countries and WHO regions.

2. Analyze temporal trends in TB cases, deaths, incidence rates, and mortality rates over the study period.

3. Identify high-burden countries and regions based on the number of TB cases and deaths.

4. Compare TB incidence and mortality rates across WHO regions and countries.

5. Assess case fatality patterns and identify countries with relatively high case fatality rates.

6. Examine the relationship between TB incidence and mortality using epidemiological data visualization and statistical analysis.

7. Develop an interactive Power BI dashboard for monitoring and exploring key TB epidemiological indicators.

8. Generate evidence-based recommendations to support TB surveillance, early detection, treatment coverage, resource allocation, and public health decision-making.

Data Source and Dataset Description

1/ Data Source

The data used in this project were obtained from the **World Health Organization (WHO) Global Tuberculosis Programme**. The WHO provides standardized tuberculosis epidemiological data to support global TB surveillance, monitoring, research, and public health decision-making.

The dataset contains country-level and regional information reported across multiple years and includes indicators related to TB disease burden, incidence, mortality, population, HIV-associated TB, and case fatality.

2/ Dataset Overview

Following data cleaning, transformation, and selection of relevant variables, the final analytical dataset contained:

Dataset Characteristic	**Description**
Data source	World Health Organization (WHO)
Disease	Tuberculosis (TB)
Observations	5,347
Analytical variables	15
Geographic level	Country
Geographic grouping	WHO Region
Time dimension	Annual observations
Output format	CSV
Analytical tools	Python and Power BI

6.3 Key Variables

The main variables used in the analysis included:

Variable	**Description**
Country	Country name
WHO_Region	WHO geographical region
Year	Year of observation
Population	Estimated population
TB_Cases	Estimated number of TB cases

TB_Deaths	Estimated number of TB deaths
TB_Incidence_Rate	Estimated TB incidence rate
Mortality_Rate	Estimated TB mortality rate
Cfr	Case fatality ratio
cfr_pct	Case fatality rate expressed as a percentage
e_tbhiv_prct	Estimated percentage of TB cases associated with HIV
iso2	Two-letter country code
ISO3	Three-letter country code
iso_numeric	Numeric country code
WHO_Region	WHO regional classification

4/ Data Preparation

The original dataset was reviewed and prepared before conducting the epidemiological analysis. Data preparation included examination of variable types, missing values, duplicate observations, consistency of country and regional classifications, and suitability of epidemiological indicators for analysis.

A cleaned analytical dataset was subsequently generated and exported as:

TB_Clean_Dataset.csv

This cleaned dataset was used as the primary input for the Power BI dashboard and subsequent visualization and reporting.

5/ Analytical Scope

The analysis focused on five major dimensions of the TB burden:

- **Disease burden:** TB cases and deaths.
- **Incidence:** TB incidence rates across countries and regions.

- **Mortality:** TB mortality rates and geographic patterns.
- **Case fatality:** Differences in the proportion of cases resulting in death.
- **Temporal and geographic variation:** Changes across years, countries, and WHO regions.

These dimensions provided the basis for the exploratory data analysis, epidemiological interpretation, and interactive Power BI dashboard developed in this project.

Methodology

1/ Study Design

This project used a descriptive epidemiological and secondary data analysis design to examine patterns of tuberculosis burden across countries and WHO regions over time.

The analysis focused on describing the distribution and temporal variation of TB cases, deaths, incidence rates, mortality rates, and case fatality rates. No individual-level patient data were analyzed.

2/ Data Analysis Workflow

The analytical workflow consisted of the following stages:

- Data acquisition
- Initial data exploration
- Data cleaning and quality assessment
- Feature engineering
- Exploratory Data Analysis (EDA)
- Epidemiological analysis
- Data visualization
- Interactive dashboard development
- Export of the cleaned dataset

The overall workflow can be summarized as:

WHO Dataset → Data Cleaning → Feature Engineering → EDA → Epidemiological Analysis → Visualization → Power BI Dashboard

3/ Data Cleaning

The dataset was first examined to understand its structure, dimensions, variable types, and data quality.

The data-cleaning process included:

- Checking the number of observations and variables.
- Reviewing variable data types.
- Identifying missing values.
- Checking for duplicate observations.
- Reviewing country and WHO region classifications.
- Converting variables to appropriate numerical or categorical formats where required.
- Reviewing epidemiological indicators for consistency.
- Selecting relevant variables for the final analytical dataset.

4/ Feature Engineering

Additional analytical indicators were derived or standardized to facilitate epidemiological interpretation.

Key derived variables included:

- Case Fatality Rate (CFR)** expressed as a percentage.
- Standardized incidence and mortality indicators.
- Aggregated annual TB cases and deaths.
- Country-level and WHO regional summaries.

5/ Exploratory Data Analysis

Exploratory Data Analysis was conducted using Python to identify patterns, trends, distributions, and relationships within the dataset.

The analysis included:

- Descriptive statistics.
- Annual aggregation of TB cases and deaths.
- Country-level comparisons.
- WHO regional comparisons.
- Analysis of incidence and mortality rates.
- Case fatality analysis.
- Correlation analysis among selected epidemiological variables.
- Identification of countries and regions with the highest TB burden.

6/ Epidemiological Analysis

The epidemiological analysis focused on several key indicators:

❖ **TB Cases**

The total number of reported or estimated TB cases was used to assess the overall disease burden across countries and years.

❖ **TB Deaths**

The number of TB deaths was analyzed to identify countries and regions experiencing the greatest mortality burden.

❖ **TB Incidence Rate**

Incidence rates were used to compare the occurrence of TB across populations with different population sizes.

❖ **Mortality Rate**

Mortality rates were assessed to examine differences in TB-related mortality across countries and WHO regions.

❖ **Case Fatality Rate**

Case Fatality Rate (CFR) was used to assess the proportion of TB cases associated with death and to identify settings with relatively high fatality.

7/ Statistical and Visualization Methods

Descriptive statistics and visual analytics were used to summarize the data.

The main visualizations included:

- Line charts for temporal trends.
- Bar charts for country-level comparisons.
- Column charts for regional comparisons.
- Scatter plots for relationships between incidence and mortality.
- Geographic maps for spatial distribution.
- Correlation matrices for relationships among selected variables.

8/ Power BI Dashboard Development

Following the Python-based analysis, the cleaned dataset was exported as a CSV file and imported into Power BI

An interactive dashboard consisting of three pages was developed:

Page 1 — Global TB Epidemiological Overview

This page presents key performance indicators, global TB trends, and an overview of the overall disease burden.

Page 2 — Country & Regional TB Burden Analysis

This page focuses on geographic variation, including high-burden countries, WHO regional comparisons, and global TB distribution.

Page 3 — TB Epidemiological Analysis

This page examines relationships between incidence, mortality, and case fatality and provides deeper epidemiological interpretation.

9/ Data Output

The final cleaned dataset was exported as:

TB_Clean_Dataset.csv

This dataset serves as the standardized analytical output and provides the data source for the Power BI dashboard.

Data Cleaning and Preparation

1/ Overview

Data cleaning and preparation were conducted to ensure that the tuberculosis dataset was suitable for epidemiological analysis and visualization. The raw dataset was systematically examined for structural inconsistencies, missing values, inappropriate data types, duplicate observations, and variables that were not required for the final analysis.

2/ Initial Data Inspection

The dataset was first inspected to understand its structure and identify the available variables.

The initial assessment included:

- Number of rows and columns.
- Variable names.
- Data types.

- Descriptive statistics.
- Missing-value patterns.
- Country and regional classifications.
- Distribution of key epidemiological indicators.

3/ Missing-Value Assessment

Missing values were assessed across all variables using Pandas.

The missing-value assessment was important because incomplete epidemiological observations can affect descriptive statistics, comparisons between countries, and the interpretation of disease indicators.

Variables with missing observations were reviewed individually to determine whether the missingness could affect the intended analysis. Where appropriate, variables with insufficient or irrelevant information were excluded from the final analytical dataset rather than introducing unsupported assumptions through arbitrary imputation.

4/ Data Type Validation

The data types of the variables were reviewed and corrected where necessary.

Variables representing:

- Years were treated as numerical or temporal variables.
- TB cases and deaths were treated as numerical variables.
- Incidence and mortality rates were treated as numerical variables.
- Country and WHO region were treated as categorical variables.
- Country identification codes were retained as categorical identifiers.

Appropriate data types ensured that calculations, aggregations, visualizations, and statistical analyses could be performed correctly.

5/ Duplicate and Consistency Checks

The dataset was reviewed for duplicate observations and inconsistencies in country and regional classifications.

Country names and WHO regional classifications were checked to ensure that observations could be grouped correctly during country-level and regional analyses.

These checks were particularly important for:

- Country comparisons.
- WHO regional aggregation.
- Geographic visualization.
- Annual trend analysis.

6/ Variable Selection

The original dataset contained a larger number of variables than were required for the main objectives of this project.

Variables directly relevant to the epidemiological analysis were retained, including:

- Country
- Country codes
- WHO Region
- Year
- Population
- TB Cases
- TB Deaths
- TB Incidence Rate
- Mortality Rate
- Case Fatality Rate
- HIV-associated TB indicators where relevant

7/ Data Transformation

Several transformations were performed to prepare the dataset for analysis.

These included:

- Standardizing variable formats.
- Creating analytical indicators where required.
- Aggregating cases and deaths by year.
- Aggregating indicators by country and WHO region.
- Preparing variables for visualization.
- Formatting case fatality measures as percentages.

8/ Final Dataset

After completing the cleaning and preparation process, the final dataset was stored as:

❖ TB_Clean_Dataset.csv

The cleaned dataset provided a consistent analytical structure for both the Python analysis and the Power BI dashboard.

The final data preparation process ensured that the subsequent findings were based on a structured and reproducible analytical dataset.

9. Feature Engineering

1/ Overview

Feature engineering was performed to transform the cleaned tuberculosis dataset into a format suitable for epidemiological analysis, comparison, and visualization. The process focused on creating and standardizing indicators that could provide meaningful measures of TB burden, severity, and geographic variation.

The derived and transformed variables were subsequently used in exploratory data analysis and the Power BI dashboard.

2/ Case Fatality Rate

The Case Fatality Rate (CFR) was used to assess the proportion of TB cases associated with death.

Where appropriate, CFR was expressed as a percentage to facilitate interpretation and comparison across countries and regions.

The resulting variable was represented as:

cfr_pct

This indicator was used to identify countries and regions with relatively high case fatality and to examine changes in fatality over time.

3/ Annual TB Cases

Annual TB cases were aggregated by year to assess temporal patterns in the global TB burden.

The resulting annual series was used to generate the:

Global TB Cases Trend Over Time

This allowed identification of years with relatively high or low numbers of TB cases.

4/ Annual TB Deaths

TB deaths were similarly aggregated by year.

The annual death series was used to assess changes in the global mortality burden and to generate the:

Global TB Mortality Trend

This provided an additional perspective on changes in TB-related mortality over time.

5/ Country-Level Aggregation

Country-level aggregations were generated to identify countries with the highest cumulative TB burden.

The analysis included:

- Total TB cases by country.

- Total TB deaths by country.
- Average incidence rate by country.
- Average mortality rate by country.
- Case fatality rate by country.

These indicators supported the identification of high-burden countries and were used in the country comparison visualizations.

6/ WHO Regional Aggregation

The dataset was also aggregated according to WHO regions to facilitate geographic comparison.

The main regional indicators included:

- Total TB cases.
- Total TB deaths.
- Average incidence rate.
- Average mortality rate.
- Case fatality rate.

Regional aggregation allowed the analysis to identify WHO regions experiencing a greater overall TB burden.

7/ Epidemiological Relationship Variables

Selected epidemiological indicators were prepared for relationship analysis.

In particular, TB incidence rate and mortality rate were examined together using a scatter plot. TB cases were incorporated as a size indicator, while WHO region was used to distinguish geographic groups.

This approach allowed the analysis to explore whether countries with higher TB incidence also tended to experience higher mortality.

8/ Summary of Engineered Features

Feature	Purpose
cfr_pct	Measure TB case fatality as a percentage
Annual TB cases	Assess temporal disease burden
Annual TB deaths	Assess temporal mortality
Country-level cases	Identify high-burden countries
Country-level deaths	Identify high-mortality countries
Regional cases	Compare TB burden by WHO region
Regional deaths	Compare mortality by WHO region
Average incidence rate	Compare disease occurrence
Average mortality rate	Compare mortality burden
Incidence–mortality relationship	Explore epidemiological patterns

9/ Analytical Output

The feature engineering process produced a structured set of epidemiological indicators that supported the subsequent **Exploratory Data Analysis, Epidemiological Findings, and Power BI Dashboard**.

These transformations improved the interpretability of the dataset and enabled consistent comparisons across countries, regions, and years.

10. Exploratory Data Analysis (EDA)

1/ Overview

Exploratory Data Analysis (EDA) was conducted to identify important patterns, distributions, trends, and relationships within the cleaned tuberculosis dataset.

The analysis was performed using Python, primarily with Pandas, NumPy, Matplotlib, and Seaborn. Descriptive statistics and graphical visualizations were used to examine the distribution of TB cases, deaths, incidence rates, mortality rates, and case fatality rates across countries, WHO regions, and years.

The EDA provided the foundation for the subsequent epidemiological interpretation and Power BI dashboard development.

2/ Descriptive Statistics

Descriptive statistics were calculated for the main epidemiological indicators.

The analysis showed the following overall values across the available observations:

Indicator	**Result**
Total TB Cases	281,399,806
Total TB Deaths	49,858,717
Average TB Incidence Rate	138.21
Average TB Mortality Rate	31.41
Average Case Fatality Rate	16.32%

These values provide an overall summary of the TB burden represented in the analytical dataset.

3/ Distribution of TB Cases

The distribution of TB cases was examined across countries and years.

The analysis identified substantial differences in cumulative TB burden between countries. India recorded the highest cumulative number of TB cases in the dataset, with approximately:

80,320,000 TB cases.

The concentration of cases among a relatively small number of high-burden countries highlights the importance of geographically targeted TB control strategies.

4/ Distribution of TB Deaths

TB deaths were analyzed to identify countries experiencing the greatest mortality burden.

India recorded the highest cumulative number of TB deaths in the analyzed dataset, with approximately:

14,151,000 deaths.

The substantial mortality burden observed in high-burden countries emphasizes the importance of early diagnosis, timely treatment initiation, treatment adherence, and effective TB surveillance.

10.5 Temporal Analysis

Annual TB cases and deaths were analyzed to examine changes over time.

The analysis covered **25 annual observations**. The highest annual number of TB cases in the analyzed series was approximately:

12,051,805 cases in 2005.

Temporal analysis provides an important perspective for understanding changes in TB burden and identifying periods requiring further epidemiological investigation.

6/ Regional Analysis

TB burden was also examined according to WHO regions.

The analysis identified the **South-East Asia (SEA)** region as having the highest cumulative number of TB cases, with approximately:

100,480,800 cases.

Regional aggregation demonstrated substantial variation in TB burden between WHO regions and provided an important basis for geographic comparison.

7/ Incidence and Mortality Analysis

TB incidence and mortality rates were examined to assess differences in disease occurrence and mortality burden across countries and regions.

The average TB incidence rate across the analyzed observations was approximately **138.21**, while the average mortality rate was approximately **31.41**.

These measures should be interpreted in relation to their respective denominators, geographic coverage, and underlying population characteristics rather than as direct measures of the number of cases or deaths.

8/ Case Fatality Analysis

Case Fatality Rate (CFR) was examined as an indicator of the proportion of TB cases associated with death.

The overall average CFR in the analytical dataset was approximately:

16.32%.

Country-level variation in CFR was explored to identify settings with relatively higher fatality and to support further epidemiological interpretation.

9/ Correlation Analysis

A correlation matrix was generated to examine relationships among selected numerical epidemiological variables.

Correlation analysis was used as an exploratory method to identify potential associations between indicators such as:

- TB cases
- TB deaths
- Population
- Incidence rate
- Mortality rate
- Case fatality rate
- Other selected epidemiological indicators

The correlation results were interpreted cautiously because correlation does not establish causation and may be influenced by population size, reporting practices, and other underlying factors.

1/0 Key Observations from EDA

The exploratory analysis identified several important patterns:

1. **TB burden is highly heterogeneous across countries**, with a substantial concentration of cases and deaths in high-burden countries.
2. **India recorded the highest cumulative TB cases and deaths** among the countries represented in the analysis.
3. **The South-East Asia region had the highest cumulative TB case burden** among the WHO regions analyzed.
4. **TB burden varied substantially over time**, with the highest annual number of cases in the analyzed series occurring in 2005.
5. **Incidence, mortality, and case fatality showed substantial geographic variation**, highlighting differences in TB epidemiological patterns across settings.
6. **The relationship between incidence and mortality requires contextual interpretation**, as mortality may be influenced by factors beyond the absolute level of TB incidence.

1/1 EDA Visualization Outputs

The exploratory analysis generated several visual outputs that were subsequently incorporated into the Power BI dashboard or used to guide the epidemiological interpretation:

- Global TB cases trend.
- Global TB deaths trend.
- Top countries by TB cases.
- Top countries by TB deaths.
- TB burden by WHO region.
- Incidence versus mortality scatter plot.

- Case fatality comparisons.
- Correlation matrix.

These visualizations provided the analytical basis for the next section, **Epidemiological Findings**, where the observed patterns are interpreted from a public health and epidemiological perspective.

Epidemiological Findings

1/ Overview

The epidemiological analysis identified substantial variation in tuberculosis (TB) burden across countries, WHO regions, and years. The findings demonstrate that TB is not distributed uniformly and that a relatively small number of countries and regions account for a large proportion of the observed disease burden.

The interpretation of these findings focuses on disease burden, mortality, temporal variation, geographic distribution, and case fatality.

2/ Global TB Disease Burden

The analysis identified approximately **281.4 million TB cases** and **49.9 million TB deaths** across the observations included in the analytical dataset.

This substantial cumulative burden demonstrates the continuing importance of tuberculosis as a global public health problem. However, cumulative totals should be interpreted carefully because the dataset represents multiple countries and years rather than a single-year global estimate.

The large difference in the number of cases and deaths also highlights the importance of examining both disease occurrence and mortality when assessing TB burden.

3/ Countries with the Highest TB Burden

India recorded the highest cumulative number of TB cases among the countries included in the analysis, with approximately **80.32 million cases**.

India also recorded the highest cumulative number of TB deaths, with approximately **14.15 million deaths**.

The concentration of a substantial proportion of the observed TB burden in high-burden countries suggests that global TB control strategies require geographically targeted approaches. Countries contributing a large share of the global burden may require sustained investment in case detection, diagnostic capacity, treatment access, adherence support, and surveillance systems.

At the same time, absolute case counts are influenced by population size. Therefore, country comparisons based on incidence and mortality rates provide complementary information and should be considered alongside absolute numbers.

4/ Regional Distribution of TB Burden

The South-East Asia (SEA) region recorded the highest cumulative TB case burden in the analyzed dataset, with approximately **100.48 million cases**.

The substantial variation between WHO regions indicates that TB control priorities differ geographically. High-burden regions may require greater emphasis on strengthening surveillance, improving diagnostic coverage, expanding treatment services, and addressing barriers to healthcare access.

Regional comparisons should nevertheless be interpreted in the context of population size, demographic characteristics, reporting systems, and differences in healthcare infrastructure.

5/ Temporal Patterns

The analysis of annual TB cases demonstrated considerable variation over the study period.

The highest annual number of TB cases in the analyzed series was approximately **12.05 million cases in 2005**.

Temporal variation in reported or estimated TB burden may reflect changes in transmission, population structure, diagnostic coverage, case detection, reporting systems, treatment programs, and other epidemiological factors.

Therefore, changes in annual case counts should not automatically be interpreted as changes in transmission alone. Additional epidemiological and contextual information is required to determine the underlying causes of observed trends.

6/ Incidence and Mortality

The average TB incidence rate across the analyzed observations was approximately **138.21**, while the average mortality rate was approximately **31.41**.

The analysis demonstrated geographic variation in both indicators. Countries or regions with elevated incidence may experience a greater need for prevention and case-finding interventions, whereas elevated mortality may indicate challenges related to delayed diagnosis, treatment access, treatment outcomes, comorbidities, or health-system capacity.

The incidence–mortality relationship was further explored using a scatter plot in the Power BI dashboard. This visualization provides a useful method for identifying countries or regions with different combinations of incidence and mortality.

7/ Case Fatality Rate

The overall average Case Fatality Rate (CFR) in the analytical dataset was approximately **16.32%**.

Variation in CFR between countries may provide an indication of differences in the severity and outcomes of TB cases. However, CFR should be interpreted cautiously because it can be affected by case detection, reporting completeness, delays in diagnosis, access to treatment, population characteristics, and differences in the underlying data.

Countries with relatively high CFR may warrant further investigation to determine whether elevated fatality is associated with delayed diagnosis, inadequate treatment coverage, comorbidities, or other health-system factors.

8/ Epidemiological Interpretation

Several important epidemiological patterns emerged from the analysis.

First, TB burden was highly concentrated geographically, with high-burden countries accounting for substantial numbers of cases and deaths.

Second, absolute disease burden and population-adjusted rates provide different perspectives. A country may have a large number of cases because of its large population, while another country may have a lower absolute burden but a substantially higher incidence or mortality rate.

Third, mortality provides an additional dimension of TB burden beyond case counts. High mortality may indicate gaps across the TB care cascade, including prevention, early detection, diagnosis, treatment initiation, treatment adherence, and management of complications.

Finally, the analysis demonstrates the importance of combining multiple epidemiological indicators rather than relying on a single measure. Cases, deaths, incidence, mortality, and CFR should be interpreted together to obtain a more comprehensive understanding of TB epidemiology.

9/ Public Health Implications

The findings have several implications for public health planning:

- High-burden countries require sustained and targeted TB control interventions.
- Early detection and diagnosis should remain central components of TB control.
- Mortality patterns should be monitored to identify potential gaps in diagnosis and treatment.
- Incidence and mortality rates should be used alongside absolute case counts for meaningful country comparisons.
- Surveillance systems should provide timely and reliable data for monitoring changes in TB burden.
- Interactive dashboards can facilitate routine monitoring and communication of epidemiological indicators.
- Data-driven resource allocation can help prioritize settings with substantial disease and mortality burdens.

Overall, the findings demonstrate the value of integrating epidemiological analysis with healthcare data analytics to support evidence-based TB surveillance and control.

Power BI Dashboard

1/ Overview

An interactive Power BI dashboard was developed to transform the results of the tuberculosis epidemiological analysis into an accessible and decision-oriented visualization tool.

The dashboard was developed using the cleaned dataset exported from the Python analysis as TB_Clean_Dataset.csv. It enables users to explore TB burden across countries, WHO regions, and years through interactive visualizations and filters.

The dashboard consists of three analytical pages, each designed to address a different aspect of tuberculosis epidemiology.

2/ Page 1 — Global TB Epidemiological Overview

The first page provides a high-level overview of the global TB burden.

Key Performance Indicators

The page includes the following key indicators:

- **Total TB Cases:** 281.4 million
- **Total TB Deaths:** 49.9 million
- **Average TB Incidence Rate:** 138.21
- **Average TB Mortality Rate:** 31.41
- **Average Case Fatality Rate:** 16.32%

Main Visualizations

The page includes:

- Global TB cases trend over time.
- Global TB mortality trend.
- TB burden by WHO region.
- Interactive filters for exploring the dataset.

This page is designed to provide a rapid overview of the overall TB burden and major temporal patterns.

3/ Page 2 — Country & Regional TB Burden Analysis

The second page focuses on geographic differences in tuberculosis burden.

Main Visualizations

The page includes:

- **Top 10 Countries by TB Cases**
- **Top 10 Countries by TB Deaths**
- **TB Burden by WHO Region**
- **Global TB Burden Map**

The geographic analysis identified **India** as the country with the highest cumulative number of TB cases and deaths in the analytical dataset.

The South-East Asia (SEA) region was identified as the WHO region with the highest cumulative TB case burden.

Interactive Filters

Users can filter the visualizations by:

- Year
- Country
- WHO Region

These interactive features allow users to investigate geographic patterns and compare TB burden across different settings.

4/ Page 3 — TB Epidemiological Analysis

The third page provides a more detailed epidemiological assessment of TB indicators and their relationships.

Main Visualizations

The page includes:

- **TB Incidence vs Mortality Scatter Plot**
- **Top 10 Countries by Case Fatality Rate**
- **Average TB Incidence Rate by WHO Region**
- **Average TB Mortality Rate by WHO Region**
- **Global Case Fatality Rate Trend**

The incidence–mortality scatter plot allows users to examine the relationship between TB incidence and mortality while incorporating TB case burden and WHO regional classification.

The case fatality analysis provides an additional perspective by highlighting countries where the proportion of cases associated with death is relatively high.

5/ Interactive Features

The dashboard incorporates interactive slicers and cross-filtering to facilitate exploratory analysis.

Users can select specific:

- Years
- Countries
- WHO regions

and observe how the selected filters affect the dashboard visualizations.

This functionality allows users to move from a global overview to more detailed country- or region-specific analysis.

6/ Dashboard Purpose

The Power BI dashboard was designed to support several public health and analytical functions, including:

- Monitoring TB disease burden.
- Identifying high-burden countries and regions.
- Examining temporal trends.
- Comparing incidence and mortality.
- Assessing case fatality patterns.
- Supporting data-driven interpretation.
- Communicating epidemiological findings to technical and non-technical audiences.

The dashboard demonstrates how epidemiological datasets can be transformed into interactive visual information that supports surveillance, reporting, and public health decision-making.

7/ Data Analytics and Public Health Value

The integration of Python-based analysis with Power BI visualization provides a reproducible analytical workflow:

WHO Data → Python Cleaning → Epidemiological Analysis → Clean CSV → Power BI Dashboard

This workflow combines data preparation, epidemiological interpretation, and interactive visualization within a single analytical project.

The resulting dashboard provides a practical example of how healthcare data analytics can complement traditional epidemiological methods and improve the communication of complex public health information.

Recommendations

Based on the epidemiological findings and patterns identified in this analysis, the following recommendations are proposed to strengthen tuberculosis prevention, surveillance, diagnosis, treatment, and public health decision-making.

1/ Strengthen Early Detection and Diagnosis

High TB burden and mortality observed in several countries highlight the importance of strengthening early detection and diagnosis.

Public health programs should:

- Expand access to TB screening and diagnostic services.
- Strengthen active case-finding strategies in high-risk populations and high-burden settings.
- Promote timely use of appropriate diagnostic technologies.
- Reduce delays between symptom onset, diagnosis, and treatment initiation.

Early identification of cases can facilitate timely treatment and reduce the risk of ongoing transmission and severe outcomes.

2/ Improve Treatment Coverage and Continuity of Care

Countries with substantial TB burden should strengthen access to effective treatment and continuity of care.

Priority actions include:

- Improving treatment coverage.
- Supporting treatment adherence.
- Reducing treatment interruption and loss to follow-up.
- Strengthening patient-centered TB services.
- Monitoring treatment outcomes through reliable surveillance systems.

Improved treatment access and continuity are essential for reducing TB-related morbidity and mortality.

3/ Prioritize High-Burden Countries and Regions

The analysis identified substantial geographic concentration of TB cases and deaths, particularly among high-burden countries and regions.

Public health resources should therefore be prioritized according to epidemiological need, with particular attention to settings experiencing:

- High numbers of TB cases.
- High mortality.
- High incidence rates.
- Relatively high case fatality rates.

Targeted resource allocation can help maximize the impact of TB prevention and control interventions.

4/ Strengthen TB Surveillance Systems

Reliable surveillance is essential for monitoring changes in TB epidemiology and evaluating the effectiveness of interventions.

Health systems should:

- Improve the completeness and timeliness of TB reporting.
- Strengthen data quality assurance and validation.
- Standardize epidemiological indicators.
- Integrate relevant surveillance platforms where appropriate.
- Promote routine analysis of TB incidence, mortality, and treatment indicators.

Improved surveillance can support earlier identification of changes in disease patterns and facilitate evidence-based decision-making.

5/ Use Data Visualization for Public Health Decision-Making

Interactive dashboards such as the Power BI dashboard developed in this project can support routine monitoring and communication of epidemiological information.

Health authorities and public health programs should consider using dashboards to:

- Monitor key TB indicators.
- Identify high-burden geographic areas.
- Track changes over time.
- Support resource allocation.
- Communicate findings to decision-makers and stakeholders.

Data visualization should complement, rather than replace, epidemiological expertise and contextual interpretation.

6/ Strengthen Integrated TB and HIV Services

Where epidemiologically appropriate, TB control programs should strengthen integration with HIV prevention, testing, treatment, and care services.

Integrated approaches can facilitate earlier identification and management of individuals at increased risk of TB-related morbidity and mortality.

7/ Promote Regular Data Updates

TB surveillance datasets should be updated regularly using the latest available WHO estimates and country-level reporting.

Regular updates are important because:

- Epidemiological patterns change over time.
- New surveillance data become available.
- Country estimates may be revised.
- Public health programs require current information for planning and evaluation.

The analytical workflow developed in this project can be reused with updated datasets to support continuous monitoring.

8/ Strengthen Data-Driven Resource Allocation

The findings demonstrate the value of combining multiple indicators rather than relying solely on absolute case numbers.

Decision-makers should consider:

- Number of TB cases.
- Number of deaths.
- Incidence rates.
- Mortality rates.
- Case fatality rates.
- Population characteristics.
- Health-system capacity.

Using multiple indicators can provide a more comprehensive basis for prioritizing interventions and allocating limited public health resources.

9/ Overall Recommendation

A comprehensive TB control strategy should combine **early detection, effective treatment, strengthened surveillance, targeted resource allocation, and data-driven decision-making**.

The integration of epidemiological expertise with healthcare data analytics can enhance the ability of public health programs to identify priorities, monitor interventions, and respond to changes in the tuberculosis burden.

Limitations

Several limitations should be considered when interpreting the findings of this tuberculosis epidemiological analysis.

1/ Secondary Data

This project relied on secondary epidemiological data obtained from the World Health Organization (WHO). The analysis therefore depends on the completeness, accuracy, consistency, and methodology of the underlying surveillance and estimation systems.

Differences in national reporting systems and data availability may influence comparisons between countries and regions.

2/ Data Completeness and Missing Values

Although the dataset was systematically assessed and cleaned, some variables contained missing observations.

Missing data may affect the calculation of averages, country-level comparisons, and the interpretation of epidemiological indicators. The analysis therefore focused on variables and observations that were sufficiently available for the intended analytical objectives.

3/ Differences in Population Size

Absolute numbers of TB cases and deaths are strongly influenced by population size.

For example, countries with large populations may have higher total numbers of cases even when their population-adjusted incidence rate is lower than that of a smaller country.

Therefore, absolute counts should be interpreted together with incidence and mortality rates.

4/ Variations in Surveillance and Reporting

Differences in national surveillance capacity, diagnostic access, reporting completeness, and case detection may influence observed TB patterns.

A lower number of reported or estimated cases does not necessarily indicate a lower underlying disease burden if detection and reporting systems are less comprehensive.

5/ Interpretation of Case Fatality Rate

Case Fatality Rate (CFR) should be interpreted cautiously.

CFR may be affected by several factors, including:

- Case detection and ascertainment.
- Delays in diagnosis.
- Access to treatment.
- Treatment outcomes.
- Population characteristics.
- Comorbidities.
- Differences in surveillance systems.

Therefore, differences in CFR between countries should not be interpreted as direct evidence of differences in treatment effectiveness without additional clinical and health-system information.

6/ Ecological Nature of the Analysis

The analysis was primarily conducted at country and regional levels rather than at the individual patient level.

Consequently, observed associations between epidemiological indicators cannot be directly interpreted as individual-level relationships. Ecological analyses may identify population-level patterns but cannot establish individual-level causation.

7/ Correlation Does Not Imply Causation

Correlation analysis was used as an exploratory technique to examine relationships between selected epidemiological indicators.

Observed correlations should not be interpreted as causal relationships. Other factors, including socioeconomic conditions, healthcare access, demographic characteristics, HIV prevalence, and health-system capacity, may influence the observed patterns.

8/ Temporal Interpretation

Changes in TB cases or deaths over time may reflect multiple factors, including changes in transmission, population structure, diagnostic practices, surveillance systems, reporting completeness, treatment programs, and data estimation methods.

Therefore, temporal trends identified in this analysis should be interpreted within their broader epidemiological and health-system context.

9/ Dashboard Limitations

The Power BI dashboard provides an interactive visualization of the available data but does not replace detailed epidemiological investigation.

The dashboard is primarily designed for descriptive analysis and monitoring. Additional statistical modeling, causal analysis, and contextual data would be required to evaluate determinants of TB burden or predict future disease trends.

10/ Generalizability

The findings describe patterns within the analyzed WHO dataset and study period. They should not automatically be generalized beyond the available data without considering differences in population characteristics, surveillance systems, healthcare infrastructure, and epidemiological context.

Despite these limitations, the analysis provides a structured and reproducible approach for exploring global TB burden and demonstrates the potential value of combining epidemiological methods with healthcare data analytics and interactive visualization.

Conclusion

Tuberculosis remains a major global public health challenge, with substantial variation in disease burden across countries, WHO regions, and years. This project applied epidemiological and healthcare data analytics methods to examine the distribution of TB cases, deaths, incidence, mortality, and case fatality using WHO data.

The analysis of the cleaned dataset identified approximately **281.4 million TB cases** and **49.9 million TB deaths** across the observations included in the dataset. India recorded the highest cumulative number of TB cases and deaths, while the South-East Asia (SEA) region recorded the highest cumulative TB case burden. The analysis also demonstrated considerable variation in incidence, mortality, and case fatality across geographic settings.

Temporal and geographic analyses provided important insights into the distribution of TB burden, while the examination of incidence and mortality demonstrated the value of using multiple epidemiological indicators to understand disease patterns. The findings also highlight the importance of interpreting absolute numbers alongside population-adjusted measures.

The project successfully integrated **Python-based data analysis** with **Power BI interactive visualization**, creating a reproducible workflow from raw epidemiological data to an interactive public health dashboard. The resulting dashboard enables users to explore TB trends, compare countries and regions, examine epidemiological relationships, and identify areas with substantial disease burden.

From a public health perspective, the findings emphasize the importance of strengthening early detection, timely diagnosis, treatment coverage, surveillance systems, and targeted allocation of resources in high-burden settings. Regular updating of epidemiological data and continued use of data-driven approaches can further support TB monitoring and control.

Overall, this project demonstrates how the integration of **epidemiology, healthcare data analytics, Python, and Power BI** can transform complex public health data into actionable information. Such approaches can support evidence-based decision-making and strengthen the capacity of public health professionals to monitor and respond to infectious disease challenges.

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Appendix C — Power BI Dashboard Screenshots

The following figures present the three interactive Power BI dashboard pages developed for this project.

Figure A1. Global TB Overview

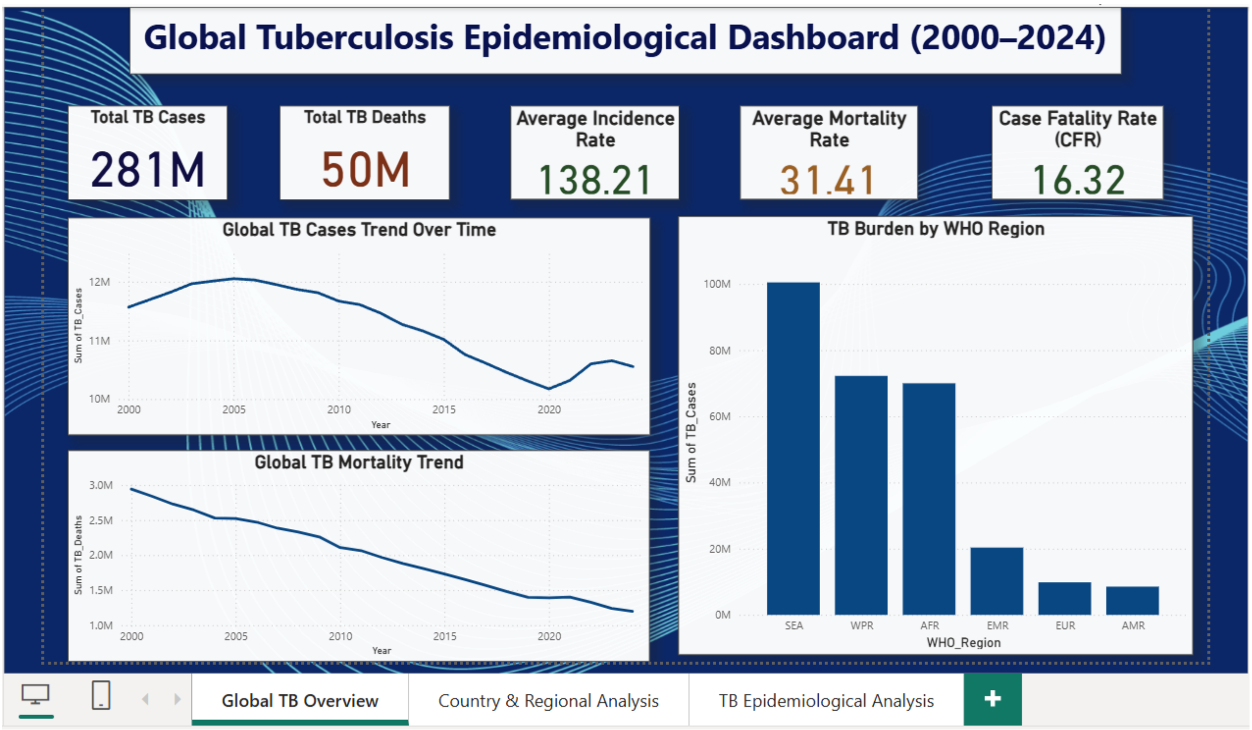

Figure A2. Country & Regional TB Burden Analysis

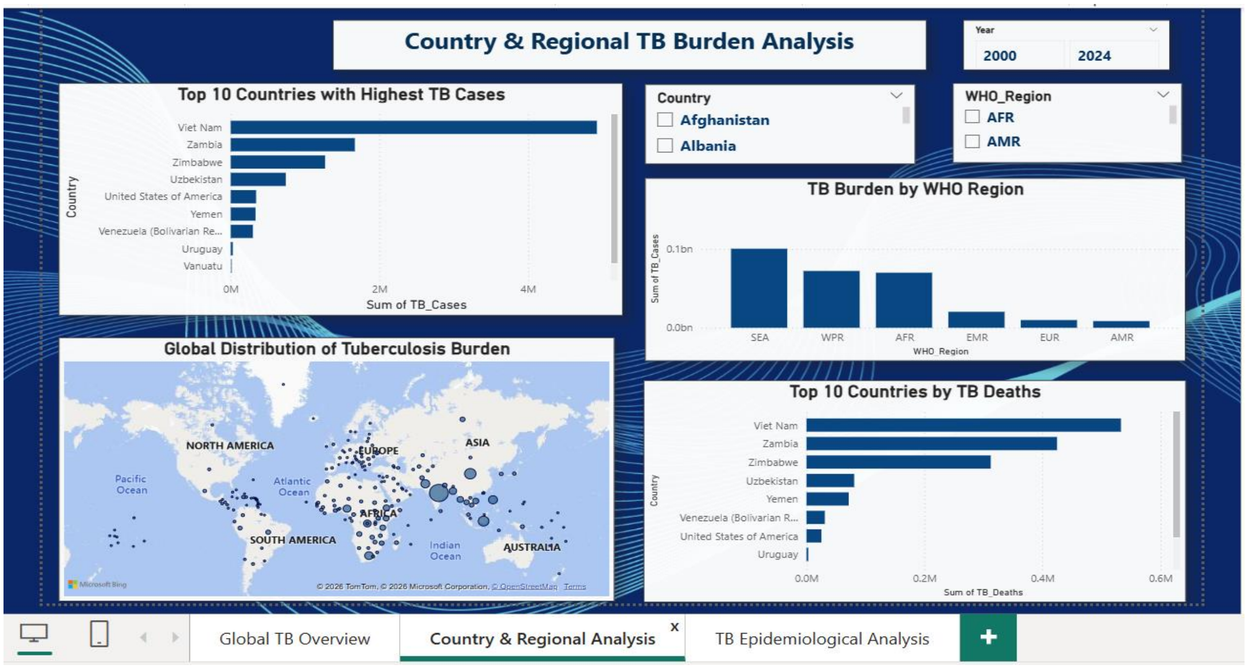

Figure A3. TB Epidemiological Analysis